

SEMESTER 1 EXAMINATIONS 2022/2023

MODULE: CA266 - Probability & Statistics

PROGRAMME(S):

COMSCI	BSc in Computer Science
CASE	BSc in Computer Applications (Sft.Eng.)
ECSAO	Study Abroad (Engineering & Computing)

YEAR OF STUDY: 2,0

EXAMINER(S):

Dr. Graham Healy (Internal) (Ext:7364)

TIME ALLOWED: 2 Hours

INSTRUCTIONS: Answer 4 questions. All questions carry equal marks.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.

The use of programmable or text storing calculators is expressly forbidden.

Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

QUESTION 1**[TOTAL MARKS: 25]**

For each of these questions **explain your answer** i.e., show your reasoning.

Q 1(a)**[12 Marks]**

Suppose you have 2 boxes each of which contains five balls. The first box contains a red ball, a blue ball, a green ball, a purple and a yellow ball. The second box contains a 2 red balls, 2 blue balls and a yellow ball.

If you draw a ball at random from the first box, and then again at random from the second box, what is the probability that:

- i. the first ball is yellow?
- ii. the second ball is yellow given the first ball was yellow?
- iii. both of the balls are not the same colour?
- iv. the first ball or the second ball or both are red?

Q 1(b)**[13 Marks]**

Suppose all the balls from both boxes in Q1(a) were combined into one box that contains 10 balls i.e., 3 red balls, 3 blue balls, 2 yellow balls, 1 purple and 1 green ball. Suppose we were to sample **without replacement** and draw N balls.

- i. What is the probability that the second ball is red given that the first ball was red?
- ii. What is the probability that the first yellow ball we get is the 5th ball?
- iii. If I draw 3 balls, what is the probability that none of these will be red or yellow?
- iv. If I draw 5 balls, what is the probability that I will observe the following sequence of ball colours: red, green, red, purple, red?

[End of Question 1]

QUESTION 2**[TOTAL MARKS: 25]**

The table below shows the number of machine failures for the past 2 years (104 weeks) for a company. For example, for 18 of the weeks there were 0 failures, for 28 of the weeks there was 1 failure, and so on.

You can assume failures are independent of each other.

Number of failures	0	1	2	3	4	5	6
Number of weeks	18	28	26	18	8	4	2

For each of these questions **explain your answer** i.e., show your reasoning.

Q 2(a)**[10 Marks]**

What are the mean and variance for the number of failures per week?

Q 2(b)**[8 Marks]**

Give values for the following probabilities:

- P(the number of failures in a week will be equal to 3)
- P(the number of failures in a week will be greater than 3)
- P(the number of failures in a week will be equal to 3 **given that** 1 failure has already occurred that week)

Q 2(c)**[7 Marks]**

The company uses the following formula to calculate the monetary cost in Euro for machine failures in a week:

$$\text{Cost} = 10X + 100$$

Where **X** = number of failures

- What is the expected mean cost of failures in a week? i.e. find $E(10X + 100)$
- What is the expected variance of cost of failures in a week? i.e. find $V(10X + 100)$

[End of Question 2]

QUESTION 3**[TOTAL MARKS: 25]**

Suppose cars on a production line have a 0.1 probability of being defective. The test used to determine if a car is defective or not is not always correct. If a car is not defective, there is a 0.01 probability that the test will show that the car is defective (positive test). If the car is defective, there is a 0.02 probability that the test will show the car is not defective (negative test).

For each of these questions **explain your answer** i.e., show your reasoning.

Q 3(a)**[3 Marks]**

What is the probability that a car is not defective?

Q 3(b)**[3 Marks]**

What is the probability of a positive test given that a car is defective?

Q 3(c)**[6 Marks]**

What is the probability that a test is positive?

Q 3(d)**[7 Marks]**

What is the probability that a car is not defective, given that a test is positive?

Q 3(e)**[6 Marks]**

What is the probability that if I select 10 random cars from the production line, that at least one of these will be defective?

[End of Question 3]

QUESTION 4**[TOTAL MARKS: 25]**

Suppose there are 2 car production lines in a factory that operate 24 hours a day, 7 days a week, 365 days a year. Both production lines are identical. Line A produces on average 1 car a day, and line B also produces on average 1 car a day.

Suppose that the number of cars produced on each production line obey a Poisson Distribution.

For each of these questions **explain your answer** i.e., show your reasoning.

Q 4(a)**[4 Marks]**

What is the probability that line A will produce 0 cars on a given day?

Q 4(b)**[5 Marks]**

What is the probability that **both** line A and line B will produce 1 or more cars **each** on a given day?

Q 4(c)**[5 Marks]**

What is the probability that in a 12-hour period, line A or line B (or both) will produce 0 cars?

Q 4(d)**[6 Marks]**

What is the probability that both production lines will produce the same number of cars on a given day? You can give your answer correct to 2 decimal places.

Q 4(e)**[5 Marks]**

For what duration of time (in hours) would the probability of line A generating no cars be 0.65?

[End of Question 4]

QUESTION 5**[TOTAL MARKS: 25]**

Answer any 2 of the following questions i.e. of Q5(a), Q5(b) or Q5(c). There are equal marks for both questions.

For each answered question **explain your answers** i.e. show your reasoning.

Q 5(a)**[12.5 Marks]**

Machine A and machine B produce the same type of component. We are told the mean length of components for both machines is 5.2cm. The standard deviation for components produced by machine A is .5 cm. We are not told the standard deviation for components produced by machine B. You can assume component lengths for both machines are normally distributed.

- i. What is the probability that a component produced by machine A has a length greater 5.7cm?
- ii. What is the probability that a component produced by machine A has a length greater than 5.7cm or less than 4.7cm?
- iii. If we are told there is a 0.242 probability that a component produced by machine B is greater than 5.7 cm, what is the standard deviation of the component lengths produced by machine B?

Q 5(b)**[12.5 Marks]**

Suppose a lowercase random 6-letter password can consist of any of the 26 letters of the alphabet.

- i. What is the probability that a random 6-letter password will not have any repeated letters?
- ii. What is the probability that a random 6-letter password will read the same backwards as forwards?

Q 5(c)**[12.5 Marks]**

A factory owner decided to secretly place GPS location tracking tags in 25 of the 400 items in a warehouse. One evening a worker decided to steal 5 random items from the warehouse.

- i. What is the probability that none of the 5 stolen items contain a tracking tag?
- ii. What is the probability that 2 or more of the stolen items contains a tracking tag?
- iii. Distinguish between the hypergeometric distribution and the binomial distribution when sampling from a finite population.

[End of Question 5]

APPENDICES

Area under the standard normal curve from $-\infty$ to z



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6338	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

${}^nC_k = \frac{{}^nP_k}{k!} = \frac{n!}{k!(n-k)!}$	
${}^nP_k = n(n-1)\dots(n-k+1) = \frac{n!}{(n-k)!}$	
$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$	
$P(A B) = \frac{P(A)P(B A)}{P(B)}$	$P(A \cap B) = \frac{P(A)P(B A)}{P(B)P(A B)}$
$V(X) = \sum_x (x - \mu_x)^2 p(x)$	$E(X) = \sum_x xp(x)$
$P(X = x) = \binom{n}{x} p^x q^{n-x}$	$P(X \leq x) = 1 - q^x$ $P(X = x) = q^{x-1} p$
$P(X = x) = \frac{\binom{M}{x} \binom{L}{n-x}}{\binom{N}{n}}$	$P(X = x) = e^{-\lambda} \frac{\lambda^x}{x!}$

[END OF APPENDICES]

[END OF EXAM]